

ICT703 – Module 10 Workshop

Learning Objectives

In this Workshop you will:

1. *Practice defining classes and objects*
2. *Practice using Pandas*

Activity 1: The Dog Park

There are lots of dogs at the dog park and we want to be able to model that using python classes.

1. Create a Dog class with an attribute species (set to "Canis familiaris") and an initialization method that assigns the name, breed and age of the dog.
2. Create 5 different dogs in the park (Park could be a list of dogs)
3. Create 2 instance methods:
 - a. description (which prints out the dogs name, age and breed) and
 - b. bark which prints out the name and the sound the dog makes (a parameter to this method – something like "woof").
4. Call the description and bark methods for your dogs in the park.
5. EXTRA – try and create a child class for each breed that barks a certain way. Inherit all the other aspects of your dog class.

Activity 2: The Movie Database

The movie database from IMDB has over 45,000 movies with details on actors, release information and ratings. We are going to work on a subset of this database "IMDB Movies Database 1" to create a pandas dataframe and work with the data.

1. Create the pandas dataframe (you can either read in the movie database or create a few lines in your program).
2. List the titles in your database.
3. Print out all the details you have on the movie "Jumanji"
4. Print out the budget and average rating of the movie "Heat"
5. Print out the average movie ratings for all movies in your dataframe
6. EXTRA – try and create a scatter plot of a movie's ratings with the budget. Use matplotlib and pandas to do this. What should you do with rows with no budget figure?